PYTHON LAB EXERCISE 14-17

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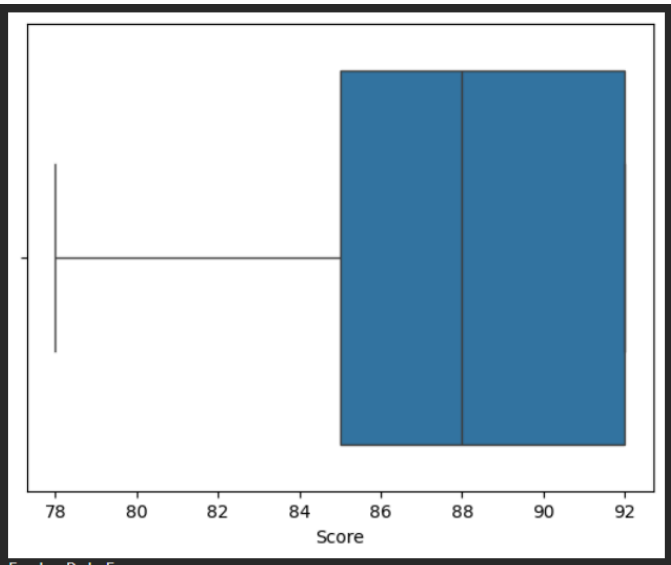
Advanced Visualization with Seaborn:
A. Using Seaborn to visualize data distributions
B. Creating a pair plot for a DataFrame

Code:

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df['Score'])
plt.show()

Q1 = df['Score'].quantile(0.25)
Q3 = df['Score'].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df['Score'] < (Q1 - 1.5 * IQR)) | (df['Score'] > (Q3 + 1.5 * IQR))]
print(outliers)
```

Output:




```
Empty DataFrame
Columns: [ID, Score, Subject, Sub_Enc, Score_Scaled]
Index: []
```

Correlation:

A. Visualize correlation matrix using heatmap from Seaborn to enhance readability

Code:

```
df_extra = pd.DataFrame({'ID': [1, 2], 'Gender': ['M', 'F']})

print(df.groupby('Subject')['Score'].mean())
merged = pd.merge(df, df_extra, on='ID', how='left')
print(merged)
```

```
Subject
Eng     88.0
Math    81.5
Sci     92.0
Name: Score, dtype: float64
   ID  Score Subject  Sub_Enc  Score_Scaled Gender
0   1     85    Math        1     -0.383482      M
1   2     92     Sci        2      0.958706      F
2   3     78    Math        1     -1.725671    NaN
3   4     92     Sci        2      0.958706    NaN
4   5     88     Eng        0      0.191741    NaN
```

Identifying Trends and Relationships:

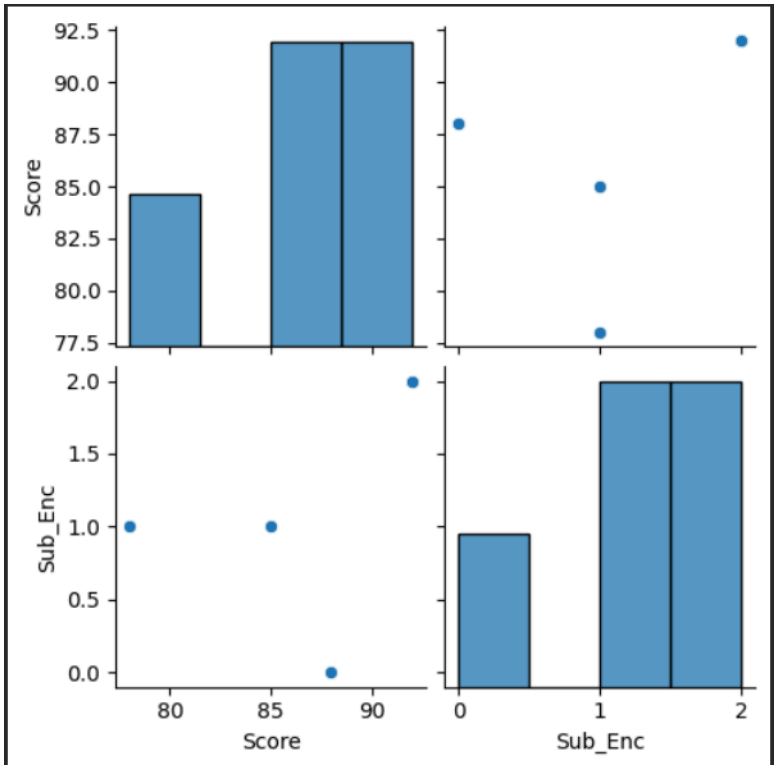
A. Identify the trend of any variable using line chart from Matplotlib and Seaborn

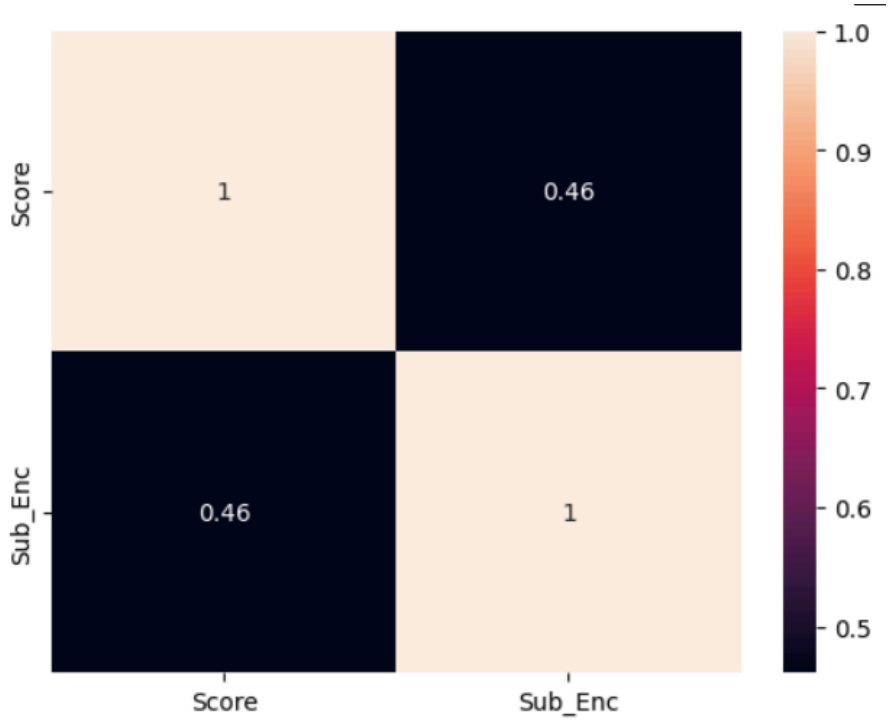
B. Visualize the relationship between two correlated columns using scatter plot

Code:

```
# Pairplot
sns.pairplot(df[['Score', 'Sub_Enc']])
plt.show()

# Heatmap
sns.heatmap(df[['Score', 'Sub_Enc']].corr(), annot=True)
plt.show()
```

Output:


92.5
90.0
87.5
85.0
82.5
80.0
77.5
Score
2.0
1.5
1.0
0.5
0.0
Sub_Enc
80
85
90
Score
0
1
2
Sub_Enc

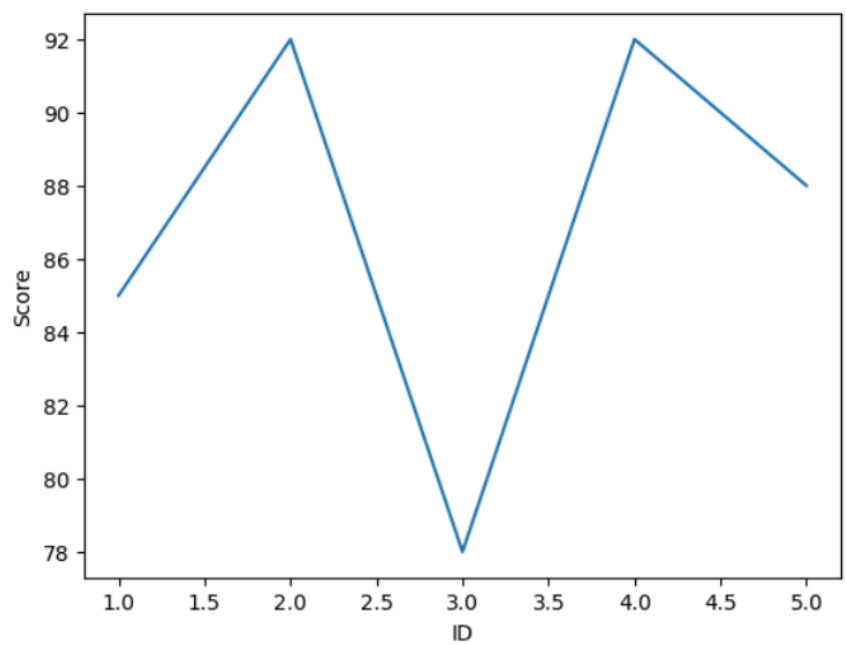

Score
Sub_Enc
1
0.46
0.46
1
Score
Sub_Enc
1.0
0.9
0.8
0.7
0.6
0.5

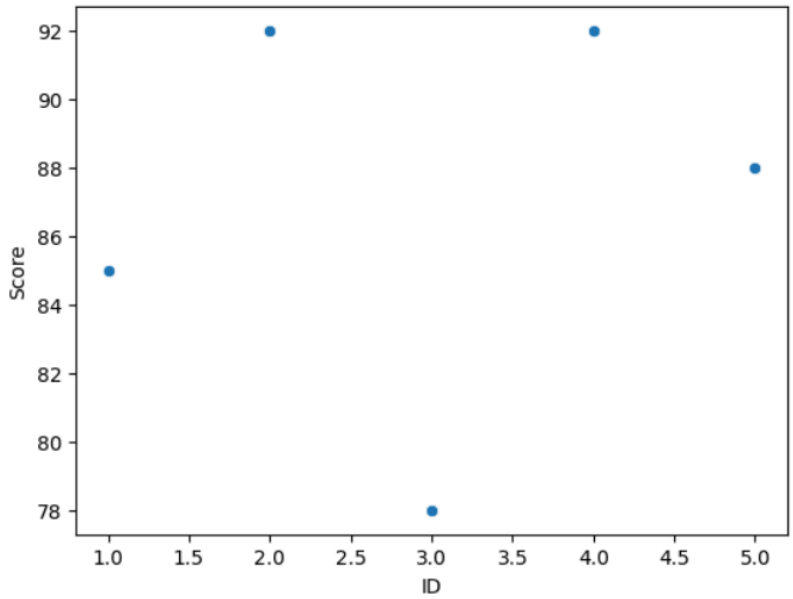
Category and Numeric Relationships:
Visualize the relationship between categorical and a numerical column using bar plot

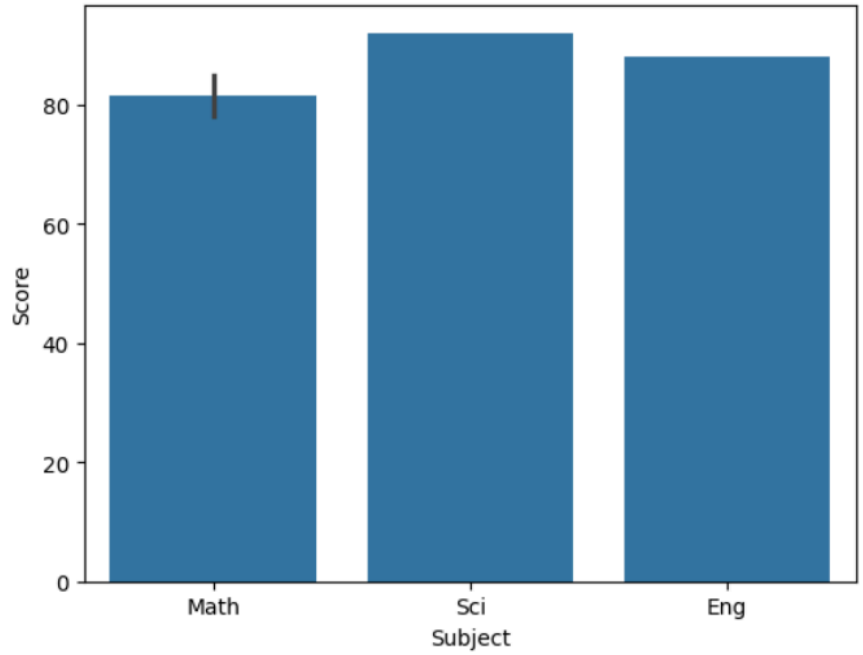
Code:

```
# Line Chart
sns.lineplot(x=df['ID'], y=df['Score'])
plt.show()

# Scatter Plot
sns.scatterplot(x='ID', y='Score', data=df)
plt.show()

# Bar Plot (Categorical vs Numerical)
sns.barplot(x='Subject', y='Score', data=df)
plt.show()
```

Output:




92
90
88
86
84
82
80
78
Score
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
ID


80
60
40
20
0
Score
Math
Sci
Eng
Subject